



Fig. 9: Wideband spectrogram of sentence "You will mark as please.."

resolving power of the spectrogram along time and frequency axes. Spectrograms obtained with N_{st} values greater than 20ms are called narrow-band spectrograms. Generally, these have a good frequency resolution. Frequency tracks appear as horizontal lines varying in intensity.

When you lower the frame-duration parameter, you need to lower the frame-shift parameter also to avoid missing frames. A reduced value of frame duration results in a lower frequency resolution but a good time resolution. The spectrogram obtained with good time resolution and poor

frequency resolution is called a wideband spectrogram. Narrowband spectrograms are useful for accurate frequency analysis of speech signals. Wideband spectrograms are useful for accurate localisation of transient region onsets in the speech signal.

Narrowband and wideband spectrograms for vowel-consonant sound 'as' are shown in Figs 6 and 7, respectively. Spectrograms (narrowband and wideband) for the sentence "You will mark as please" are shown in Figs 8 and 9, respectively. Readers are advised to experiment with different speech utterances and different values for N_{st} , N_{sf} , and $nfft$ and to observe variations in spectrograms. **EFY**



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[Development background]

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● When it is assumed the LED lighting is only the life of capacitor

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